

Covariance of windowed bispectrum multipoles: a guided derivation

A step-by-step companion to `bispectrum_covariance.typ`, with its corrections folded in

Branch `thecov2`, October 2026. Conventions of `pk_covariance.typ` (power spectrum) and of Sugiyama, Saito, Beutler & Seo (2019, 2020) for the bispectrum basis.

Road map. The derivation has the same skeleton as the power-spectrum one; each step is one idea.

1. **Geometry.** The estimator is a triangle anchored at the galaxy that carries the line of sight, with two legs $\mathbf{r}_1, \mathbf{r}_2$ ending at the galaxies whose momenta are binned (Section 2).
2. **Locality.** Every two-point correlator is a window at one point times a locally anisotropic power spectrum; connected correlators collapse to a point (Section 3).
3. **Wick.** Of the 15 Gaussian pairings of six fields, six connect the two estimators; one more survives on the $k_1 = k_2$ elements (Section 4).
4. **Plane waves on the legs.** In each pairing the phases recombine into one plane wave per leg of the triangle (Section 5).
5. **Shell averages.** Averaging the momenta of one estimator over their shells moves its angular weight from the momenta onto the legs, at the cost of bin-averaged Bessel functions. The other estimator brings the power spectra, including that of the unbinned third side (Section 5.3).
6. **Rotational invariance.** What multiplies the window is a rotational scalar of five directions. So only a few invariant projections of the window's three-point function are needed (Section 5.4), and these are computed from pair sums without triplets (Section 8).

What is new relative to $P(k)$: a three-point window function, the third side $k_3 = |\mathbf{k}_1 + \mathbf{k}_2|$ that is not binned, and a “ $k_3 \approx 0$ ” term on the $k_1 = k_2$ elements. The non-Gaussian terms (PB, BB, PT) turn out to be **simpler**: they reduce to the power-spectrum formula with other window weights (Section 10).

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1 Notation and three tools

Fourier and harmonics. $\tilde{f}(\mathbf{k}) = \int d^3x e^{-i\mathbf{k}\cdot\mathbf{x}} f(\mathbf{x})$; orthonormal $Y_{\ell m}$ with $\bar{Y}_{\ell m} = (-1)^m Y_{\ell, -m}$; Legendre polynomials $\mathcal{L}_\ell$. All estimator and power-spectrum multipoles are even.

Shell averages. For a radial bin i (with all directions),

$$\langle\langle f \rangle\rangle_i \equiv \frac{\int_i q^2 dq \int d\Omega_q / (4\pi) f(q)}{\int_i q^2 dq}, \quad V_i \equiv \int_i \frac{d^3 q}{(2\pi)^3} = \frac{1}{2\pi^2} \int_i q^2 dq. \quad (1)$$

Nested averages over several bins are written $\langle\langle f \rangle\rangle_{ij}$ and so on. Radial bin averages of Bessel functions carry a bar:

$$\bar{j}_\lambda^{(i)}(r) \equiv \frac{\int_i q^2 dq j_\lambda(qr)}{\int_i q^2 dq}, \quad \bar{g}_\lambda^{(i)}(r) \equiv \frac{\int_i q^2 dq j_\lambda(qr) g(q)}{\int_i q^2 dq}. \quad (2)$$

Tool 1 (addition theorem). $\mathcal{L}_\ell(\hat{\mathbf{a}} \cdot \hat{\mathbf{b}}) = 4\pi/(2\ell+1) \sum_m Y_{\ell m}(\hat{\mathbf{a}}) \bar{Y}_{\ell m}(\hat{\mathbf{b}})$. The conjugate may sit on either factor.

Tool 2 (plane wave over a shell, i.e. Funk–Hecke). For any function h_ℓ in the span of the $Y_{\ell m}$ (for example $Y_{\ell m}$ or $\bar{Y}_{\ell m}$),

$$\langle\langle e^{\pm i\mathbf{q}\cdot\mathbf{r}} h_\ell(\hat{\mathbf{q}}) g(q) \rangle\rangle_i = (\pm i)^\ell \bar{g}_\ell^{(i)}(r) h_\ell(\hat{\mathbf{r}}). \quad (3)$$

A harmonic weight on the momentum reappears unchanged on the separation. This is the workhorse of the whole derivation; with $g = 1$ it gives $\bar{j}_\ell^{(i)}$.

Tool 3 (opening-angle expansion). For a function of the opening angle, $f(\hat{\mathbf{p}}_1 \cdot \hat{\mathbf{p}}_2) = \sum_a f_a \mathcal{L}_a(\hat{\mathbf{p}}_1 \cdot \hat{\mathbf{p}}_2)$, two applications of Equation 3 and Tool 1 give

$$\int \frac{d\Omega_1 d\Omega_2}{(4\pi)^2} e^{i\mathbf{p}_1 \cdot \mathbf{r}_1} e^{\pm i\mathbf{p}_2 \cdot \mathbf{r}_2} \mathcal{L}_a(\hat{\mathbf{p}}_1 \cdot \hat{\mathbf{p}}_2) = (\mp 1)^a j_a(p_1 r_1) j_a(p_2 r_2) \mathcal{L}_a(\hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2), \quad (4)$$

that is, $(-1)^a$ for two plane waves of the same sign and $+1$ for opposite signs. (Both signs were checked by Monte Carlo.)

The Sugiyama basis. With $y_\ell^m = \sqrt{4\pi/(2\ell+1)} Y_{\ell m}$, $H_{\ell_1 \ell_2 L} = \begin{pmatrix} \ell_1 & \ell_2 & L \\ 0 & 0 & 0 \end{pmatrix}$ and $N_{\ell_1 \ell_2 L} = (2\ell_1+1)(2\ell_2+1)(2L+1)$, the multipoles are defined by

$$B(\mathbf{k}_1, \mathbf{k}_2; \hat{\mathbf{n}}) = \sum_\ell B_\ell(k_1, k_2) \mathcal{S}_\ell(\hat{\mathbf{k}}_1, \hat{\mathbf{k}}_2, \hat{\mathbf{n}}), \quad (5)$$

$$\mathcal{S}_\ell(\hat{\mathbf{a}}, \hat{\mathbf{b}}, \hat{\mathbf{n}}) = \frac{1}{H_\ell} \sum_{m_1 m_2 M} \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & M \end{pmatrix} y_{\ell_1}^{m_1}(\hat{\mathbf{a}}) y_{\ell_2}^{m_2}(\hat{\mathbf{b}}) y_L^M(\hat{\mathbf{n}}),$$

with $\ell \equiv (\ell_1 \ell_2 L)$. The 3- j symbol makes $\mathcal{S}_\ell$ a rotational scalar of its three arguments. It is real when $\ell_1 + \ell_2 + L$ is even; we keep the bar $\bar{\mathcal{S}}$ anyway, for bookkeeping. Two things are used repeatedly:

- **Orthogonality.** At fixed $\hat{\mathbf{n}}$, averaging $\bar{\mathcal{S}}_\ell \mathcal{S}_{\ell'}$ over $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ gives $\delta_{\ell\ell'}/(N_\ell H_\ell^2)$.
- **Unnormalised version.** $\tilde{\mathcal{S}}_\ell \equiv \sum \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & M \end{pmatrix} Y_{\ell_1 m_1} Y_{\ell_2 m_2} Y_{LM}$, so that $\mathcal{S}_\ell = (4\pi)^{3/2}/(H_\ell \sqrt{N_\ell}) \tilde{\mathcal{S}}_\ell$.

Examples: $\mathcal{S}_{000} = 1$. $H_{\ell 0 \ell} = (-1)^\ell / \sqrt{2\ell+1}$, so $H_{202} = +1/\sqrt{5}$, $N_{202} = 25$ and $\mathcal{S}_{202}(\hat{\mathbf{a}}, \hat{\mathbf{b}}, \hat{\mathbf{n}}) = \mathcal{L}_2(\hat{\mathbf{a}} \cdot \hat{\mathbf{n}})$. (All of these were checked numerically.)

Fields and windows. One tracer, $F(\mathbf{x}) = w(\mathbf{x})[n_g(\mathbf{x}) - \alpha n_r(\mathbf{x})]$, real, so $F(-\mathbf{k}) = \overline{F(\mathbf{k})}$. With $\omega = \bar{n}\langle w \rangle$ the mean weighted density (`weight_average.typ`),

$$W = \omega^2 \text{ (pair window)}, \quad S = (1 + \alpha) \bar{n} \langle w^2 \rangle \text{ (pair shot noise)}, \quad I_3 = \int d^3 x \omega^3 \text{ (normalisation)}. \quad (6)$$

2 The estimator is a triangle anchored at the line of sight

Sugiyama’s estimator puts the line of sight on one galaxy, at $\mathbf{x}$, and bins the momenta $\mathbf{q}_1, \mathbf{q}_2$ of the other two:

$$\hat{B}_\ell(i, j) = \frac{(4\pi)^{3/2} H_\ell \sqrt{N_\ell}}{I_3} \left\langle \left\langle \int d^3 x \bar{\mathcal{S}}_\ell(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{x}}) e^{i(\mathbf{q}_1 + \mathbf{q}_2) \cdot \mathbf{x}} F(\mathbf{q}_1) F(\mathbf{q}_2) F(\mathbf{x}) \right\rangle \right\rangle_{ij}. \quad (7)$$

Writing the two Fourier fields out,

$$e^{i(\mathbf{q}_1 + \mathbf{q}_2) \cdot \mathbf{x}} F(\mathbf{q}_1) F(\mathbf{q}_2) = \int d^3 y_1 d^3 y_2 e^{-i\mathbf{q}_1 \cdot (\mathbf{y}_1 - \mathbf{x})} e^{-i\mathbf{q}_2 \cdot (\mathbf{y}_2 - \mathbf{x})} F(\mathbf{y}_1) F(\mathbf{y}_2). \quad (8)$$

Each binned momentum is thus conjugate to one **leg** of the triangle, $\mathbf{r}_a = \mathbf{y}_a - \mathbf{x}$, measured from the anchor.

Warm-up: the mean. Assume the three-point function is local, $\langle F(\mathbf{y}_1) F(\mathbf{y}_2) F(\mathbf{x}) \rangle \approx \omega^3(\mathbf{x}) \zeta(\mathbf{r}_1, \mathbf{r}_2; \hat{\mathbf{x}})$, i.e. the window is constant over the triangle. The legs then integrate to the bispectrum $B(\mathbf{q}_1, \mathbf{q}_2; \hat{\mathbf{x}})$ (third momentum $-\mathbf{q}_1 - \mathbf{q}_2$), and

$$\langle \hat{B}_\ell \rangle = \frac{(4\pi)^{3/2} H_\ell \sqrt{N_\ell}}{I_3} \int d^3 x \omega^3(\mathbf{x}) \left\langle \left\langle \bar{\mathcal{S}}_\ell B \right\rangle \right\rangle_{ij} = \frac{\int \omega^3}{I_3} \langle \langle B_\ell \rangle \rangle_{ij}, \quad (9)$$

by orthogonality. This fixes the prefactor of Equation 7. The window dilutes the multipole by $\int \omega^3 / I_3 = 1$; a pair-averaged window would replace this by the bispectrum analogue of I_k / norm .

3 The local approximation: three rules

We use the same approximations as for the power spectrum:

- **(A1)** The window is constant over a correlation length.
- **(A2)** Nearby points share the line of sight.
- **(A3)** Connected correlators are confined within a correlation length.

Under A1–A2, $\langle F(\mathbf{y}) F(\mathbf{y} + \mathbf{u}) \rangle \approx W(\mathbf{y}) \xi(\mathbf{u}; \hat{\mathbf{y}})$ (plus $S(\mathbf{y}) \delta_D(\mathbf{u})$), with $P(\mathbf{p}; \hat{\mathbf{n}}) = \sum_L \mathcal{L}_L(\hat{\mathbf{p}} \cdot \hat{\mathbf{n}}) P_L(p)$ its transform. In the covariance the fields appear in three guises: Fourier fields, configuration fields integrated against an estimator phase, and the anchor. This gives three rules, each a one-line change of variables $\mathbf{u} = (\text{separation})$ followed by $\int d^3 u e^{-i\mathbf{p} \cdot \mathbf{u}} \xi(\mathbf{u}) = P(\mathbf{p})$:

$$\begin{aligned} \text{(R1)} \quad \langle F(\mathbf{q}) F(-\mathbf{p}) \rangle &\approx \int d^3 y e^{-i(\mathbf{q} - \mathbf{p}) \cdot \mathbf{y}} W(\mathbf{y}) P(\mathbf{p}; \hat{\mathbf{y}}), \\ \text{(R2)} \quad \int d^3 x' e^{-i\mathbf{p} \cdot \mathbf{x}'} \langle F(\mathbf{q}) F(\mathbf{x}') \rangle &\approx \int d^3 y e^{-i(\mathbf{q} + \mathbf{p}) \cdot \mathbf{y}} W(\mathbf{y}) P(\mathbf{p}; \hat{\mathbf{y}}), \\ \text{(R3)} \quad \int d^3 x' e^{-i\mathbf{p} \cdot \mathbf{x}'} \langle F(\mathbf{x}) F(\mathbf{x}') \rangle &\approx e^{-i\mathbf{p} \cdot \mathbf{x}} W(\mathbf{x}) P(\mathbf{p}; \hat{\mathbf{x}}). \end{aligned} \quad (10)$$

Reading the rules.

- A correlator is the window **at one point** times the anisotropic spectrum with the line of sight **at that point**.
- The spectrum is evaluated at the momentum of the partner field, conventionally the primed one. Choosing the other one, or the mean, changes the result at $O(\Delta k/k)$.
- When a field that carries an estimator's line-of-sight harmonic is absorbed into a correlator anchored elsewhere, the harmonic moves to that anchor (A2).

Shot noise is restored at the end by $WP_L \rightarrow WP_L + S \delta_{L0}^K$ in each correlator: everything is linear in these window–spectrum pairs.

4 Wick: which pairings survive

The covariance is $C = \langle \hat{B} \overline{\hat{B}'} \rangle - \langle \hat{B} \rangle \langle \overline{\hat{B}'} \rangle$. The primed estimator has shells (i', j') , multipoles $\ell' = (\ell'_1 \ell'_2 L')$ and momenta $\mathbf{p}_1, \mathbf{p}_2$; conjugating Equation 7 turns its phase into $e^{-i(\mathbf{p}_1 + \mathbf{p}_2) \cdot \mathbf{x}'}$ and its fields into $F(-\mathbf{p}_1)F(-\mathbf{p}_2)F(\mathbf{x}')$. Label the six fields

$$1 = F(\mathbf{q}_1), \quad 2 = F(\mathbf{q}_2), \quad 3 = F(\mathbf{x}), \quad 1' = F(-\mathbf{p}_1), \quad 2' = F(-\mathbf{p}_2), \quad 3' = F(\mathbf{x}'). \quad (11)$$

A Gaussian six-point function is a sum over the 15 ways of pairing six fields. Sorting them:

pairings	what the momenta must satisfy	status
class 1: $(11')(22')(33')$, $(12')(21')(33')$	$\mathbf{q}_a \approx \mathbf{p}_b$: leg to leg	leading
class 2: $(11')(23')(32')$ and 3 relabellings	a leg of one triangle equals the <i>third</i> <i>side</i> of the other	$O(\Delta k/k)$ on the diagonal
$(12)(1'2')(33')$	$\mathbf{q}_1 + \mathbf{q}_2 \approx 0$, $\mathbf{p}_1 + \mathbf{p}_2 \approx 0$, i.e. $k_3, k'_3 \approx 0$	on $i = j$, $i' = j'$ only (Section 7)
the other 8 with a pair inside one estimator	a shell momentum below the window scale k_w	vanish

For example, (13) forces $\mathbf{q}_2 \approx 0$ after the $\mathbf{x}$ integral. Only $(12)(1'2')(33')$ can be satisfied with all momenta in their shells, by the folded triangles with $k_3 \rightarrow 0$ that exist when $i = j$.

5 Class 1: the two anchors pair together

5.1 Phases become plane waves on the legs

For $(11')(22')(33')$ use R1 for $(11')$ at $\mathbf{y}_1$ and for $(22')$ at $\mathbf{y}_2$, and R3 with $\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2$ for $(33')$, multiplied by the unprimed phase $e^{i(\mathbf{q}_1 + \mathbf{q}_2) \cdot \mathbf{x}}$. The phases combine as

$$e^{-i(\mathbf{q}_1 - \mathbf{p}_1) \cdot \mathbf{y}_1} e^{-i(\mathbf{q}_2 - \mathbf{p}_2) \cdot \mathbf{y}_2} e^{i(\mathbf{q}_1 + \mathbf{q}_2 - \mathbf{p}_1 - \mathbf{p}_2) \cdot \mathbf{x}} = e^{-i(\mathbf{q}_1 - \mathbf{p}_1) \cdot \mathbf{r}_1} e^{-i(\mathbf{q}_2 - \mathbf{p}_2) \cdot \mathbf{r}_2}. \quad (12)$$

Each momentum pair rides a plane wave on one leg. The three windows sit at the three vertices of the triangle:

$$C_{\ell\ell'}^{(1)} = \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2} \quad (13)$$

$$\times \int d^3x d^3r_1 d^3r_2 W(\mathbf{x}) W(\mathbf{x} + \mathbf{r}_1) W(\mathbf{x} + \mathbf{r}_2) \mathcal{K}^{(1)}(\mathbf{r}_1, \mathbf{r}_2; \hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2, \hat{\mathbf{x}}),$$

$$\mathcal{K}^{(1)} = \left\langle \left\langle e^{-i\mathbf{q}_1 \cdot \mathbf{r}_1 - i\mathbf{q}_2 \cdot \mathbf{r}_2} \tilde{\mathcal{S}}_\ell(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{x}}) \right\rangle \right\rangle_{ij} \quad (14)$$

$$\times \left\langle \left\langle e^{i\mathbf{p}_1 \cdot \mathbf{r}_1 + i\mathbf{p}_2 \cdot \mathbf{r}_2} \tilde{\mathcal{S}}_{\ell'}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \hat{\mathbf{x}}) P(\mathbf{p}_1; \hat{\mathbf{y}}_1) P(\mathbf{p}_2; \hat{\mathbf{y}}_2) P(|\mathbf{p}_1 + \mathbf{p}_2|; \hat{\mathbf{x}}) \right\rangle \right\rangle_{i'j'},$$

with $\hat{\mathbf{y}}_a = (\mathbf{x} + \mathbf{r}_a) / |\mathbf{x} + \mathbf{r}_a|$. The double shell average has **factorised** into one average per estimator. This is the analogue of the “one plane wave per binned momentum” step of the power-spectrum note.

The second class-1 pairing $(12')(21')(33')$ is the same with $(\mathbf{p}_1, \ell'_1, i') \leftrightarrow (\mathbf{p}_2, \ell'_2, j')$. It needs $i = j'$ and $j = i'$, so on a diagonal element $(i, j; i, j)$ it contributes only if $i = j$, and then with equal weight.

Check: the periodic box. Set $W = 1$, $\hat{\mathbf{x}} = \hat{\mathbf{y}}_a = \hat{\mathbf{n}}$, $I_3 = V$. The $\mathbf{r}_a$ integrals give $(2\pi)^3 \delta_D(\mathbf{q}_a - \mathbf{p}_a)$, whose shell average is $\delta_{ii'}/V_i$:

$$\begin{aligned}
C_{\text{box}}^{(1)} &= \frac{(4\pi)^3 H H' \sqrt{N N'}}{V V_i V_j} \delta_{ii'} \delta_{jj'} \left\langle \langle \bar{\mathcal{S}}_\ell \tilde{\mathcal{S}}_{\ell'} P(\mathbf{q}_1) P(\mathbf{q}_2) P(|\mathbf{q}_1 + \mathbf{q}_2|) \rangle \right\rangle_{ij} \\
&= \frac{N N' H^2 H'^2 V}{N_{\text{mode}(k_i)} N_{\text{mode}(k_j)}} \delta_{ii'} \delta_{jj'} \left\langle \langle \bar{\mathcal{S}}_\ell \mathcal{S}_{\ell'} P P P \rangle \right\rangle_{ij},
\end{aligned} \tag{15}$$

with $N_{\text{mode}(k_i)} = V V_i$. This is the first term of eq. (49) of Sugiyama et al. (2020); the “5 perms.” there are the other five pairings.

5.2 Warm-up: B_{000} in real space

Everything essential is visible in the simplest case. Here $\tilde{\mathcal{S}}_{000} = (4\pi)^{-3/2}$, $H = N = 1$, and $P(\mathbf{p}; \hat{\mathbf{n}}) = P(p)$. In $\mathcal{K}^{(1)}$:

1. **Unprimed average.** By Equation 3 with $\ell = 0$, $\langle \langle e^{-i\mathbf{q}_1 \cdot \mathbf{r}_1} \rangle \rangle_i = \bar{j}_0^{(i)}(r_1)$, and the same for leg 2.
2. **Primed average.** The only coupling between the legs is the third side. Expand it in the opening angle, $P(|\mathbf{p}_1 + \mathbf{p}_2|) = \sum_a p_a(p_1, p_2) \mathcal{L}_a(\hat{\mathbf{p}}_1 \cdot \hat{\mathbf{p}}_2)$ with $p_a = (2a + 1)/2 \int_{-1}^1 d\mu \mathcal{L}_a(\mu) P(\sqrt{p_1^2 + p_2^2 + 2p_1 p_2 \mu})$. Tool 3 (same signs) gives

$$\langle \langle \dots \rangle \rangle_{i'j'} = \sum_a (-1)^a \mathcal{P}_a^{(i'j')}(r_1, r_2) \mathcal{L}_a(\hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2), \tag{16}$$

$$\mathcal{P}_a^{(i'j')}(r_1, r_2) \equiv \frac{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2 j_a(p_1 r_1) j_a(p_2 r_2) P(p_1) P(p_2) p_a(p_1, p_2)}{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2}. \tag{17}$$

3. **Window.** What remains of the window is its three-point function projected on the opening angle of the triangle:

$$C_{000}^{(1)} = \frac{1}{I_3^2} \sum_a (-1)^a \int r_1^2 dr_1 \int r_2^2 dr_2 \bar{j}_0^{(i)}(r_1) \bar{j}_0^{(j)}(r_2) \mathcal{P}_a^{(i'j')}(r_1, r_2) \mathcal{Q}_a(r_1, r_2), \tag{18}$$

$$\mathcal{Q}_a(r_1, r_2) \equiv \int d^3x \int d\Omega_{r_1} d\Omega_{r_2} W(\mathbf{x}) W(\mathbf{x} + \mathbf{r}_1) W(\mathbf{x} + \mathbf{r}_2) \mathcal{L}_a(\hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2). \tag{19}$$

In the box, $\mathcal{Q}_a = (4\pi)^2 V \delta_{a0}$. With $4\pi \int r^2 dr \bar{j}_0^{(i)}(r) \bar{g}_0^{(i')}(r) = \delta_{ii'} \langle \langle g \rangle \rangle_i / V_i$, Equation 18 becomes $\delta_{ii'} \delta_{jj'} \langle \langle P P p_0 \rangle \rangle_{ij} / (V V_i V_j)$: the box result, in which only the **angle-averaged** third side p_0 survives. A window feeds the higher Legendre orders a of the third side into the covariance, in proportion to the anisotropy of its three-point function, $\mathcal{Q}_{a>0}$. The rest of this section only adds indices to Equation 18.

5.3 General multipoles: the two shell averages

Unprimed estimator. $\bar{\mathcal{S}}_\ell$ is a product of $\bar{Y}_{\ell_1 m_1}(\hat{\mathbf{q}}_1) \bar{Y}_{\ell_2 m_2}(\hat{\mathbf{q}}_2)$ and a harmonic of $\hat{\mathbf{x}}$, and nothing else in its average depends on $\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2$. Tool 2 applied twice gives the exact statement

$$\left\langle \langle e^{-i\mathbf{q}_1 \cdot \mathbf{r}_1 - i\mathbf{q}_2 \cdot \mathbf{r}_2} \bar{\mathcal{S}}_\ell(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{x}}) \rangle \right\rangle_{ij} = (-i)^{\ell_1 + \ell_2} \bar{j}_{\ell_1}^{(i)}(r_1) \bar{j}_{\ell_2}^{(j)}(r_2) \bar{\mathcal{S}}_\ell(\hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2, \hat{\mathbf{x}}). \tag{20}$$

The estimator’s angular weight is simply transferred from the momenta to the legs.

Primed estimator. Here three power spectra multiply the weight. Expand the two leg spectra with Tool 1, putting the conjugates on the momenta,

$$P(\mathbf{p}_a; \hat{\mathbf{y}}_a) = \sum_{L_a} \frac{4\pi}{2L_a + 1} \sum_{M_a} \bar{Y}_{L_a M_a}(\hat{\mathbf{p}}_a) Y_{L_a M_a}(\hat{\mathbf{y}}_a) P_{L_a}(p_a), \tag{21}$$

and do the same for the vertex, $P(\mathbf{p}_3; \hat{\mathbf{x}}) = \sum_{L_3} 4\pi / (2L_3 + 1) \sum_{\gamma} \bar{Y}_{L_3 \gamma}(\hat{\mathbf{p}}_3) Y_{L_3 \gamma}(\hat{\mathbf{x}}) P_{L_3}(p_3)$.

The vertex is the one new ingredient. $\bar{Y}_{L_3\gamma}(\hat{\mathbf{p}}_3)P_{L_3}(|\mathbf{p}_1 + \mathbf{p}_2|)$ is a function of both momenta, and it transforms as a rank- L_3 object. Its expansion in bipolar harmonics, the tensor generalisation of the opening-angle expansion in Section 5.2, is

$$\bar{Y}_{L_3\gamma}(\hat{\mathbf{p}}_3)P_{L_3}(p_3) = \sum_{ab} c_{ab}^{L_3}(p_1, p_2) \sum_{\alpha\beta} \langle a\alpha b\beta \mid L_3\gamma \rangle \bar{Y}_{a\alpha}(\hat{\mathbf{p}}_1) \bar{Y}_{b\beta}(\hat{\mathbf{p}}_2). \quad (22)$$

The coefficients $c_{ab}^{L_3}$ are real and independent of γ . Each is a one-dimensional integral over the opening angle. Parity requires $a + b + L_3$ even and $|a - b| \leq L_3 \leq a + b$. For $L_3 = 0$ they reduce to the Legendre coefficients p_a of Section 5.2, up to normalisation. A few tens of orders resolve the BAO in the third side.

Each primed momentum now carries a plane wave and three harmonics: those of $\tilde{\mathcal{S}}_{\ell'}$, of the leg spectrum, and of the vertex. Tool 2 moves their product to the leg. Let

$$g_{\ell L a \Lambda}^{m M \alpha \nu} \equiv \int d\Omega Y_{\ell m} \bar{Y}_{L M} \bar{Y}_{a \alpha} \bar{Y}_{\Lambda \nu} \quad (23)$$

be the four-harmonic Gaunt integral that re-expands the product as harmonics of rank Λ , with $\Lambda \leq \ell + L + a$. The two radial averages **do not** factorise, because $c_{ab}^{L_3}$ depends on both magnitudes. They form one two-dimensional kernel:

$$\mathcal{P}_{L_1 L_2 L_3; ab; \Lambda \Lambda'}^{(i' j')}(r_1, r_2) \equiv \frac{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2 j_{\Lambda}(p_1 r_1) j_{\Lambda'}(p_2 r_2) P_{L_1}(p_1) P_{L_2}(p_2) c_{ab}^{L_3}(p_1, p_2)}{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2}. \quad (24)$$

The primed average is then

$$\sum_{L_1 L_2 L_3} \frac{(4\pi)^3}{(2L_1 + 1)(2L_2 + 1)(2L_3 + 1)} \sum_{ab} \sum_{\Lambda \Lambda'} i^{\Lambda + \Lambda'} \mathcal{P}_{L_1 L_2 L_3; ab; \Lambda \Lambda'}^{(i' j')}(r_1, r_2) \mathcal{A}(\hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2, \hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2, \hat{\mathbf{x}}), \quad (25)$$

where $\mathcal{A}$ is a fixed polynomial in harmonics:

$$\begin{aligned} \mathcal{A} = & \sum \left(\begin{matrix} \ell'_1 & \ell'_2 & L' \\ m'_1 & m'_2 & M' \end{matrix} \right) \langle a\alpha b\beta \mid L_3\gamma \rangle g_{\ell'_1 L_1 a \Lambda}^{m'_1 M_1 \alpha \nu_1} g_{\ell'_2 L_2 b \Lambda'}^{m'_2 M_2 \beta \nu_2} \\ & \times Y_{\Lambda \nu_1}(\hat{\mathbf{r}}_1) Y_{\Lambda' \nu_2}(\hat{\mathbf{r}}_2) Y_{L_1 M_1}(\hat{\mathbf{y}}_1) Y_{L_2 M_2}(\hat{\mathbf{y}}_2) Y_{L' M'}(\hat{\mathbf{x}}) Y_{L_3 \gamma}(\hat{\mathbf{x}}). \end{aligned} \quad (26)$$

5.4 Rotational invariance: why the window enters through a few numbers

Collect the result: $\mathcal{K}^{(1)}$ is a sum of products of radial functions (of r_1, r_2) and angular functions of the five directions $\hat{\mathbf{r}}_1, \hat{\mathbf{y}}_1, \hat{\mathbf{r}}_2, \hat{\mathbf{y}}_2, \hat{\mathbf{x}}$. Every ingredient is built from dot products, so $\mathcal{K}^{(1)}$ is unchanged when all five directions are rotated together. A function of five directions with this property lies in the span of the **invariant** functions

$$\begin{aligned} \Phi_n(\hat{\mathbf{r}}_1, \hat{\mathbf{y}}_1, \hat{\mathbf{r}}_2, \hat{\mathbf{y}}_2, \hat{\mathbf{x}}) = & \sum \left(\begin{matrix} K_1 & K_2 & \Lambda_3 \\ \kappa_1 & \kappa_2 & \mu_3 \end{matrix} \right) \\ & \times [Y_{\lambda_1}(\hat{\mathbf{r}}_1) \otimes Y_{L_1}(\hat{\mathbf{y}}_1)]_{K_1 \kappa_1} [Y_{\lambda_2}(\hat{\mathbf{r}}_2) \otimes Y_{L_2}(\hat{\mathbf{y}}_2)]_{K_2 \kappa_2} Y_{\Lambda_3 \mu_3}(\hat{\mathbf{x}}), \end{aligned} \quad (27)$$

where $[Y_{\lambda} \otimes Y_L]_{K\kappa} = \sum \langle \lambda \rho L M \mid K \kappa \rangle Y_{\lambda \rho} Y_{L M}$, and $n = (\lambda_1 L_1 K_1; \lambda_2 L_2 K_2; \Lambda_3)$. The coupling tree follows the geometry: leg 1, leg 2, and the anchor. The Φ_n are orthonormal on the five spheres. So

$$\mathcal{K}^{(1)} = \sum_n \mathcal{K}_n(r_1, r_2) \Phi_n, \quad \mathcal{K}_n = \int d^5\Omega \bar{\Phi}_n \mathcal{K}^{(1)}. \quad (28)$$

The physical directions are not independent ($\hat{\mathbf{y}}_a$ is fixed by $\mathbf{x}$ and $\mathbf{r}_a$), but this identity holds for independent directions, so it holds in particular for the physical ones. Inserting it in Equation 13, the window is needed only through its projections on the same functions. This is the Wigner–Eckart step:

$$C_{\ell\ell'}^{(1)}(ij; i'j') = \frac{(4\pi)^3 HH' \sqrt{NN'}}{I_3^2} \sum_n \int r_1^2 dr_1 \int r_2^2 dr_2 \mathcal{K}_n(r_1, r_2) \mathcal{Q}_n(r_1, r_2), \quad (29)$$

$$\mathcal{Q}_n(r_1, r_2) \equiv \int d^3x \int d\Omega_{r_1} d\Omega_{r_2} W(\mathbf{x}) W(\mathbf{x} + \mathbf{r}_1) W(\mathbf{x} + \mathbf{r}_2) \Phi_n(\hat{\mathbf{r}}_1, \hat{\mathbf{y}}_1, \hat{\mathbf{r}}_2, \hat{\mathbf{y}}_2, \hat{\mathbf{x}}), \quad (30)$$

$$\begin{aligned} \mathcal{K}_n &= (-i)^{\ell_1 + \ell_2} \bar{j}_{\ell_1}^{(i)}(r_1) \bar{j}_{\ell_2}^{(j)}(r_2) \\ &\times \sum_{L_3 ab\Lambda\Lambda'} \frac{(4\pi)^3 i^{\Lambda + \Lambda'}}{(2L_1 + 1)(2L_2 + 1)(2L_3 + 1)} \mathcal{P}_{L_1 L_2 L_3; ab; \Lambda\Lambda'}^{(i'j')}(r_1, r_2) T_n, \end{aligned} \quad (31)$$

$$T_n \equiv \int d^5\Omega \bar{\Phi}_n \bar{\mathcal{S}}_\ell(\hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2, \hat{\mathbf{x}}) \mathcal{A} \quad (\text{a pure number, from 3-j algebra or quadrature}). \quad (32)$$

The L_1, L_2 of $\mathcal{K}_n$ are the ranks of n at $\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2$. Three kinds of object appear, cleanly separated:

- **Window** ($\mathcal{Q}_n$): geometry only.
- **Radial kernels** ($\bar{j}, \mathcal{P}$): cosmology and binning.
- **Coupling numbers** (T_n): angular-momentum algebra, computed once.

Selection rules.

- $\lambda_1 \in \text{tri}(\ell_1, \Lambda)$ and $\lambda_2 \in \text{tri}(\ell_2, \Lambda')$ (merging $\bar{Y}_{\ell_a}$ with Y_Λ on the leg).
- $\Lambda \leq \ell'_1 + L_1 + a$ and $\Lambda' \leq \ell'_2 + L_2 + b$.
- Λ_3 even, $\Lambda_3 \leq L + L' + L_3$.
- K_a runs over **all** integers $|\lambda_a - L_a| \leq K_a \leq \lambda_a + L_a$. Two different directions are coupled here, so there is no parity restriction; for example, $(\hat{\mathbf{r}}_1 \times \hat{\mathbf{y}}_1) \cdot (\hat{\mathbf{r}}_2 \times \hat{\mathbf{y}}_2)$ is a legitimate invariant.

The phase $(-i)^{\ell_1 + \ell_2} i^{\Lambda + \Lambda'}$ is real, because $\Lambda + \Lambda' \equiv a + b \equiv L_3 \equiv 0 \pmod{2}$.

Check: the periodic box. With $W = 1$ and a global line of sight, the leg integrals $\int d\Omega_{r_a} Y_{\lambda_a \rho_a}(\hat{\mathbf{r}}_a)$ force $\lambda_1 = \lambda_2 = 0$, hence $K_a = L_a$. Using $\sum_{M_1 M_2 \mu_3} \begin{pmatrix} L_1 & L_2 & \Lambda_3 \\ M_1 & M_2 & \mu_3 \end{pmatrix} Y_{L_1 M_1} Y_{L_2 M_2} Y_{\Lambda_3 \mu_3}(\hat{\mathbf{n}}) = H_{L_1 L_2 \Lambda_3} \sqrt{N_{L_1 L_2 \Lambda_3}} / (4\pi)^{3/2}$ (checked numerically),

$$\mathcal{Q}_n^{\text{box}} = V \sqrt{N_{L_1 L_2 \Lambda_3}} / 4\pi H_{L_1 L_2 \Lambda_3} \delta_{\lambda_1 0} \delta_{\lambda_2 0}. \quad (33)$$

Then $\Lambda = \ell_1$ and $\Lambda' = \ell_2$, and Equation 29 reduces to the box formula above. The box therefore tests the kernels, the counting factors and T_n independently of any pair count. This was done during the review with 4000 Gaussian boxes, for B_{000} , B_{202} and B_{220} .

5.5 Truncation is set by the window, not by the kernel

The vertex sums over (a, b) do not terminate. But a kernel term of rank λ_1 at $\hat{\mathbf{r}}_1$ only survives if $\mathcal{Q}_n$ has that rank. The window varies on scales of hundreds of $h^{-1}\text{Mpc}$, so the high-rank projections $\mathcal{Q}_n$ are small. In a box only $\lambda_a = 0$ survives. Keeping $\lambda_1, \lambda_2 \leq \lambda_w$ bounds every sum:

$$\Lambda \leq \lambda_w + \ell_1, \quad a \leq \lambda_w + \ell_1 + \ell'_1 + L_1, \quad b \leq \lambda_w + \ell_2 + \ell'_2 + L_2. \quad (34)$$

Convergence in λ_w is checked by computing $\mathcal{Q}_n$ to higher rank, at pair-count cost, and against Gaussian realisations on the footprint. The truncation error is set by the window's angular smoothness on the scales $r \sim 1/\Delta k$ that dominate the near-diagonal elements.

5.6 Shot noise

Each of the three correlators is linear in its window–spectrum pair. Replacing $WP_L \rightarrow WP_L + S\delta_{L0}^K$ at the two legs and the anchor gives eight terms:

- A shot-noise **leg** has $L_a = 0$ and $P_{L_a} \rightarrow 1$ in Equation 24.
- A shot-noise **anchor** has $L_3 = 0$ and $c_{ab}^0 \rightarrow \sqrt{4\pi} \delta_{a0} \delta_{b0}$.

The window functions become $\mathcal{Q}_n^{XYZ}$ with $X, Y, Z \in \{W, S\}$.

6 Class 2: an anchor pairs with a leg

Take (11')(23')(32'). R1 gives (11') at $\mathbf{y}_1$. R2 with $\mathbf{p} = \mathbf{p}_3 \equiv \mathbf{p}_1 + \mathbf{p}_2$ gives (23') at $\mathbf{y}_2$, where the primed line of sight now sits. R1 read backwards gives $\int d^3\mathbf{y}' e^{i\mathbf{p}_2 \cdot \mathbf{y}'} \langle F(\mathbf{x}) F(\mathbf{y}') \rangle = e^{i\mathbf{p}_2 \cdot \mathbf{x}} W(\mathbf{x}) P(\mathbf{p}_2; \hat{\mathbf{x}})$. The phases are

$$e^{-i(\mathbf{q}_1 - \mathbf{p}_1) \cdot \mathbf{r}_1} e^{-i(\mathbf{q}_2 + \mathbf{p}_3) \cdot \mathbf{r}_2}. \quad (35)$$

Change variables from $\mathbf{p}_2$ to $\mathbf{p}_3$. The Jacobian is 1, and the shell condition on $\mathbf{p}_2$ becomes the indicator $\mathbb{1}[\mathbf{p}_3 - \mathbf{p}_1 \in j']$, while $\mathbf{p}_3$ runs over all magnitudes. Now each leg again carries exactly one primed momentum ($\mathbf{p}_1$ on leg 1, $\mathbf{p}_3$ on leg 2), and the structure is that of class 1:

$$C_{\ell\ell'}^{(2)} = \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2} \times \int d^3x d^3r_1 d^3r_2 W(\mathbf{x}) W(\mathbf{x} + \mathbf{r}_1) W(\mathbf{x} + \mathbf{r}_2) \mathcal{K}^{(2)}(\mathbf{r}_1, \mathbf{r}_2; \hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2, \hat{\mathbf{x}}), \quad (36)$$

$$\mathcal{K}^{(2)} = (-i)^{\ell_1 + \ell_2} \bar{j}_{\ell_1}^{(i)}(r_1) \bar{j}_{\ell_2}^{(j)}(r_2) \bar{\mathcal{S}}_\ell(\hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2, \hat{\mathbf{x}}) \times \left\langle \left\langle e^{i\mathbf{p}_1 \cdot \mathbf{r}_1 - i\mathbf{p}_3 \cdot \mathbf{r}_2} \bar{\mathcal{S}}_{\ell'}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \hat{\mathbf{y}}_2) P(\mathbf{p}_1; \hat{\mathbf{y}}_1) P(\mathbf{p}_3; \hat{\mathbf{y}}_2) P(\mathbf{p}_2; \hat{\mathbf{x}}) \right\rangle \right\rangle_{i', \mathbf{p}_3}, \quad (37)$$

where the last average runs over $\mathbf{p}_1 \in i'$ and $\mathbf{p}_3$ with $|\mathbf{p}_3 - \mathbf{p}_1| \in j'$, normalised by $\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2$. Everything that depends on $\mathbf{p}_2 = \mathbf{p}_3 - \mathbf{p}_1$ forms the **class-2 vertex**: the indicator, $Y_{\ell'_2}(\hat{\mathbf{p}}_2)$ from $\bar{\mathcal{S}}_{\ell'}$, and $P(\mathbf{p}_2; \hat{\mathbf{x}})$. It is expanded in bipolar harmonics of $(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_3)$ exactly as in Equation 22.

- The five directions are the same as in class 1, so the **same invariant window functions** $\mathcal{Q}_n$ appear. Only the rank assignment differs: $\hat{\mathbf{y}}_2$ carries the primed line of sight coupled with the leg-2 spectrum, and $\hat{\mathbf{x}}$ carries the unprimed line of sight with the vertex spectrum.
- In real space, B_{000} gives Equation 18 with $(-1)^a \rightarrow +1$ (Tool 3, opposite signs) and $\mathcal{P}_a \rightarrow \mathcal{P}_a^{(2)}$. Here $j_a(p_3 r_2)$ is integrated over all p_3 , and p_a is replaced by the Legendre coefficients of $\mathbb{1}[\mathbf{p}_3 - \mathbf{p}_1 \in j'] P(|\mathbf{p}_3 - \mathbf{p}_1|)$ in $\hat{\mathbf{p}}_1 \cdot \hat{\mathbf{p}}_3$.
- The other three class-2 pairings follow by $(\mathbf{p}_1, \ell'_1, i') \leftrightarrow (\mathbf{p}_2, \ell'_2, j')$ and $(\mathbf{q}_1, \ell_1, i, \mathbf{r}_1) \leftrightarrow (\mathbf{q}_2, \ell_2, j, \mathbf{r}_2)$.

Check: the box, and why class 2 is small on the diagonal. The leg integrals force $\mathbf{q}_1 = \mathbf{p}_1$ and $\mathbf{q}_2 = -\mathbf{p}_3$; the shell of $\mathbf{q}_2$ must contain the **third side** of the primed triangle,

$$C_{\text{box}}^{(2)} = \frac{(4\pi)^3 H H' \sqrt{N N'}}{V V_i V_j} \delta_{ii'} \times \left\langle \left\langle \mathbb{1}[\mathbf{p}_1 + \mathbf{p}_2 \in j] \bar{\mathcal{S}}_\ell(\hat{\mathbf{p}}_1, -\hat{\mathbf{p}}_3, \hat{\mathbf{n}}) \bar{\mathcal{S}}_{\ell'}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \hat{\mathbf{n}}) P(p_1) P(p_2) P(p_3) \right\rangle \right\rangle_{ij'}. \quad (38)$$

Since $p_3^2 = p_1^2 + p_2^2 + 2p_1 p_2 \mu$, the indicator keeps a fraction $\Delta k p_3 / (2p_1 p_2) \approx \Delta k / (2k)$ of the opening angles. Counting the pairings that survive (two of class 2 against one of class 1 for $i \neq j$; four against two for $i = j$), class 2 is $\approx \Delta k / k$ of class 1 on the diagonal, i.e. 5% at $k = 0.1$ with $\Delta k = 0.005$. Off the diagonal it is the only Gaussian contribution.

7 The $k_3 \approx 0$ term

For (12)(1'2')(33'), R1 gives $\langle F(\mathbf{q}_1) F(\mathbf{q}_2) \rangle = \int d^3\mathbf{y} e^{-i\boldsymbol{\varepsilon} \cdot \mathbf{y}} W(\mathbf{y}) P(\mathbf{q}_2; \hat{\mathbf{y}})$ with $\boldsymbol{\varepsilon} = \mathbf{q}_1 + \mathbf{q}_2$. This needs $|\boldsymbol{\varepsilon}| \lesssim k_w$, i.e. folded triangles with $\mathbf{q}_2 \approx -\mathbf{q}_1$, which exist only if $i = j$. The average over the $\mathbf{q}_2$ shell

turns $\int d^3\varepsilon/(2\pi)^3 e^{i\varepsilon \cdot (\mathbf{x}-\mathbf{y})}$ into $\delta_D(\mathbf{x}-\mathbf{y})/V_i$. The same happens on the primed side, and the anchors remain connected by (33'):

$$C^{(7)}(ii; i' i') = \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2 V_i V_{i'}} \times \int d^3x d^3x' W(\mathbf{x}) W(\mathbf{x}') \left\langle \langle \bar{\mathcal{S}}_\ell(\hat{\mathbf{q}}, -\hat{\mathbf{q}}, \hat{\mathbf{x}}) P(\mathbf{q}; \hat{\mathbf{x}}) \rangle \right\rangle_i \times \left\langle \langle \bar{\mathcal{S}}_{\ell'}(\hat{\mathbf{p}}, -\hat{\mathbf{p}}, \hat{\mathbf{x}}') P(\mathbf{p}; \hat{\mathbf{x}}') \rangle \right\rangle_{i'} \langle F(\mathbf{x}) F(\mathbf{x}') \rangle_{\text{IC}}. \quad (39)$$

What this is. On the $k_1 = k_2$ elements the Sugiyama multipole includes the triangles with $k_3 \rightarrow 0$, where $\hat{B}$ is the local power $P(k)$ times the W -weighted mean overdensity of the survey.

- In a box with a fixed mean the latter vanishes (the integral constraint, “IC”).
- In the survey, α forces $\int F = 0$ but not $\int \omega^2 F = 0$, so the term is zero for uniform ω .
- It reached tens of percent of the (i, i) variance in a Gaussian test with a radially varying ω .

It has the same mode-counting factor $1/(V_i V_{i'})$ as class 1, so it is not suppressed by $\Delta k/k$.

8 The window invariants from pair sums

The coupling tree of Φ_n was chosen so that $\mathcal{Q}_n$ factorises at each anchor:

$$\mathcal{Q}_n(r_1, r_2) = \int d^3x W(\mathbf{x}) \sum_{\kappa_1 \kappa_2 \mu_3} \begin{pmatrix} K_1 & K_2 & \Lambda_3 \\ \kappa_1 & \kappa_2 & \mu_3 \end{pmatrix} Y_{\Lambda_3 \mu_3}(\hat{\mathbf{x}}) A_{K_1 \kappa_1}^{\lambda_1 L_1}(\mathbf{x}; r_1) A_{K_2 \kappa_2}^{\lambda_2 L_2}(\mathbf{x}; r_2), \quad (40)$$

$$A_{K\kappa}^{\lambda L}(\mathbf{x}; r) \equiv \int d\Omega_r W(\mathbf{x} + \mathbf{r}) [Y_\lambda(\hat{\mathbf{r}}) \otimes Y_L(\hat{\mathbf{y}})]_{K\kappa}. \quad (41)$$

A is a **pair** quantity: the harmonic moments of the window on the sphere of radius r about the anchor. It is accumulated from the pairs (anchor, neighbour) in bins of r , exactly like the pair counts of `WindowLibrary`. The three-point function is the sum over anchors of products of two such moments, so no triplets are counted (the Slepian–Eisenstein construction). Practical points:

- **Frame.** Rotate each anchor’s frame so that $\hat{\mathbf{x}} = \hat{\mathbf{z}}$. Then $Y_{\Lambda_3 \mu_3}(\hat{\mathbf{x}}) \propto \delta_{\mu_3 0}$, which removes one magnetic sum.
- **Self-pairs.** When r_1 and r_2 fall in the same bin, the product AA contains the terms with the same neighbour in both factors. These have zero measure in the continuum and must be subtracted.
- **Cost.** With 5000 anchors and the neighbours of the power-spectrum pair counts, the moments cost as now. The products cost about 10^{12} operations for $\lambda_w = 4$, $L_a \leq 2$ on a 100×100 radial grid. The near-pair regime uses the tree neighbours as now.
- **Weights.** The eight shot-noise combinations need the moments with W or S at the neighbour and the anchor.

9 Worked index sets: B_{000} and B_{202}

B_{000} . $\tilde{\mathcal{S}}_{000} = (4\pi)^{-3/2}$ on both sides cancels the $(4\pi)^3$ of the prefactor. Leg 1 forces $\lambda_1 = \Lambda \in \text{tri}(L_1, a)$ and leg 2 forces $\lambda_2 = \Lambda' \in \text{tri}(L_2, b)$; at the anchor $\Lambda_3 = L_3$. In real space this is Equation 18. In redshift space $L_{1,2,3} \in \{0, 2, 4\}$: the line-of-sight ranks of the window function at the two far vertices and at the anchor are those of the two leg spectra and of the vertex spectrum.

B_{202} . Here $H = 1/\sqrt{5}$, $N = 25$, $\mathcal{S}_{202} = \mathcal{L}_2(\hat{\mathbf{k}}_1 \cdot \hat{\mathbf{n}})$, and by Equation 20 the unprimed weight becomes $-\bar{j}_2^{(i)}(r_1) \bar{j}_0^{(j)}(r_2) \mathcal{L}_2(\hat{\mathbf{r}}_1 \cdot \hat{\mathbf{x}})$ up to normalisation: a quadrupole of leg 1 about the anchor’s line of sight. Then:

- Leg 2 still has $\lambda_2 = \Lambda'$.

- Leg 1 has $\lambda_1 \in \text{tri}(2, \Lambda)$ with $\Lambda \leq 2 + L_1 + a$.
- At the anchor, Λ_3 is even and $\Lambda_3 \leq 4 + L_3$.

Since $\ell_1 \neq \ell_2$, the second class-1 pairing is not the mirror of the first and must be evaluated separately (it contributes on $i = j$).

10 Non-Gaussian terms: back to the power-spectrum formula

The localisation rule (A3). A connected correlator of three fields is confined to one point:

$$\langle F(\mathbf{k}_a)F(\mathbf{k}_b)F(\mathbf{k}_c) \rangle_c \approx \int d^3y e^{-i(\mathbf{k}_a+\mathbf{k}_b+\mathbf{k}_c)\cdot\mathbf{y}} \omega^3(\mathbf{y}) B(\mathbf{k}_a, \mathbf{k}_b; \hat{\mathbf{y}}). \quad (42)$$

For four fields the same holds with ω^4 and the trispectrum. A configuration field with estimator phase $e^{i\mathbf{K}\cdot\mathbf{x}}$ acts as a Fourier field of momentum $-\mathbf{K}$ at $\mathbf{x}$. If the correlator contains the anchor, there is no $\mathbf{y}$ integral and $\omega^3(\mathbf{x})$ appears.

Why these terms are simpler than the Gaussian one. PB is a pair plus a connected triple, BB is two connected triples, and PT is a pair plus a connected quadruple. In each there are exactly **two points**, so there is **one separation $\mathbf{s}$** and one plane wave. The window enters through a pair count with weights (ω^3, W) , (ω^3, ω^3) or (ω^4, W) , i.e. the functions $\mathcal{Q}_{\Lambda_1\Lambda_2\Lambda}^{\omega\omega'}(\mathbf{s})$ of the power-spectrum note with other weights. There is no three-point window function. The momenta that do not carry the plane wave are averaged over their shells **before** anything else, and they turn the bispectrum or trispectrum into an **effective spectrum**. What is left is literally the power-spectrum shell kernel $\mathcal{K}_{\ell L}^{(i)}[\mathbf{p}](\hat{\mathbf{a}}, \hat{\mathbf{b}}; \mathbf{s})$ of `pk_covariance.typ` (its eq. “kernel-def”), and its final formula applies.

One more approximation is needed, at the order already accepted:

- **(A4)** Inside slowly varying factors, set the beat momentum ($\varepsilon =$ the sum of the two momenta tied by the plane wave, $|\varepsilon| \lesssim k_w$) to zero.

This is the same $O(k_w/k)$ freedom as “which momentum carries the spectrum”. For PT it is exactly the separation from super-sample covariance (below).

10.1 PB

With $\hat{P}_\ell(i) = (2\ell + 1)/I_2 \langle \langle \int d^3x e^{-i\mathbf{k}\cdot\mathbf{x}} \mathcal{L}_\ell(\hat{\mathbf{k}} \cdot \hat{\mathbf{x}}) F(\mathbf{x}) F(-\mathbf{k}) \rangle \rangle_i$ and $I_2 = \int \omega^2$, label $a = F(-\mathbf{k})$ and $b = F(\mathbf{x})$. Five fields have no Gaussian part. The leading term pairs one field of $\hat{P}$ with one of $\hat{B}'$ and connects the other three. The pairings inside $\hat{B}'$ are beat modes and (ab) is disconnected. That leaves six:

pairing	points: pair / triple	plane wave	box coincidence
$(a1')(b2'3')$, mirror $1' \leftrightarrow 2'$	$W(\mathbf{x} + \mathbf{s}) / \omega^3(\mathbf{x})$	$e^{i(\mathbf{k}+\mathbf{p}_1)\cdot\mathbf{s}}$	$k \approx p_1$: full weight
$(b1')(a2'3')$, mirror	$W(\mathbf{x}) / \omega^3(\mathbf{x} + \mathbf{s})$	$e^{i(\mathbf{k}-\mathbf{p}_1)\cdot\mathbf{s}}$	$k \approx p_1$: full weight
$(a3')(b1'2')$, $(b3')(a1'2')$	as above	carries $\mathbf{k}$ and $\mathbf{p}_3$	$k \approx p'_3$: $O(\Delta k/k)$

For $(a1')(b2'3')$, R1 gives the pair and the localisation rule gives the triple, anchored at $\mathbf{x}$, $\omega^3(\mathbf{x})e^{-i\mathbf{p}_1\cdot\mathbf{x}}B(-\mathbf{p}_2, \mathbf{p}_1 + \mathbf{p}_2; \hat{\mathbf{x}})$, with the $\hat{B}'$ line of sight moved to $\hat{\mathbf{x}}$. Only $\mathbf{k}$ and $\mathbf{p}_1$ carry the plane wave, so average over $\mathbf{p}_2$ first:

$$\beta(\mathbf{p}_1; \hat{\mathbf{x}}) \equiv \langle \langle \tilde{\mathcal{S}}_{\ell'}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \hat{\mathbf{x}}) B(-\mathbf{p}_2, \mathbf{p}_1 + \mathbf{p}_2; \hat{\mathbf{x}}) \rangle \rangle_{j'} = \sum_J \beta_J(p_1) \mathcal{L}_J(\hat{\mathbf{p}}_1 \cdot \hat{\mathbf{x}}), \quad (43)$$

a Legendre series because it is a rotational scalar of two directions. Then

$$\begin{aligned}
C_{\ell,\ell'}^{PB,(a1')}(i;i'j') &= \frac{(2\ell+1)(4\pi)^{3/2}H'\sqrt{N'}}{I_2I_3} \sum_{L_1J} \\
&\times \int d^3x d^3s \omega^3(\mathbf{x})W(\mathbf{x}+\mathbf{s}) \\
&\times \mathcal{K}_{\ell 0}^{(i)}[1](\hat{\mathbf{x}}, \cdot; \mathbf{s}) \mathcal{K}_{JL_1}^{(i')}[P_{L_1}\beta_J](\hat{\mathbf{x}}, \hat{\mathbf{y}}; \mathbf{s}),
\end{aligned} \tag{44}$$

with $\hat{\mathbf{y}} = (\mathbf{x} + \mathbf{s})/|\mathbf{x} + \mathbf{s}|$. This is the power-spectrum master formula with weights (ω^3, W) and effective spectrum $P_{L_1}\beta_J$. In the box it gives $\delta_{ii'}/V_i$ times $P(k)B(k, \mathbf{p}_2, |\mathbf{k} - \mathbf{p}_2|)$, averaged with the weights: the PB term of Sugiyama et al. (2020).

The pairings $(b1')(a2'3')$ are the same with the weights swapped, $W(\mathbf{x})$ and $\omega^3(\mathbf{x} + \mathbf{s})$, and the $\hat{B}'$ line of sight at $\hat{\mathbf{y}}$. In the third-side pairings use $(\mathbf{p}_1, \mathbf{p}_3)$: the plane-wave momentum is $\mathbf{p}_3$, integrated over all magnitudes with the indicator, which is where the $\Delta k/k$ comes from.

10.2 BB

The six fields split into two connected triples in 10 ways. $(123)(1'2'3')$ is $\langle \hat{B} \rangle \langle \hat{B}' \rangle$; the other nine each join one unprimed field to two primed ones. The triple containing the unprimed singleton u and the primed pair (a', b') forces $k_u \approx k_{c'}$, where c' is the remaining primed field:

singleton pair	coincidence	weight on the diagonal $(i, j) = (i', j')$
$(1 2'3'), (2 1'3')$	$k_1 \approx k'_1, k_2 \approx k'_2$	full
$(1 1'3'), (2 2'3')$	$k_1 \approx k'_2, k_2 \approx k'_1$	full if $i = j$, else zero
$(3 1'3'), (3 2'3'), (1 1'2'), (2 1'2'), (3 1'2')$	involve a third side	$O(\Delta k/k)$

Take $(1 | 2'3')$: the triple $(1, 2', 3')$ is anchored at $\mathbf{x}'$ and the triple $(2, 3, 1')$ at $\mathbf{x}$. Their phases give $e^{-i(\mathbf{q}_1 + \mathbf{p}_1) \cdot \mathbf{s}}$ with $\mathbf{s} = \mathbf{x}' - \mathbf{x}$, so $\mathbf{q}_1 \approx -\mathbf{p}_1$. Each estimator's line of sight stays at its own anchor. With A4 the two bispectra become the unprimed and primed triangles, and the averages over the non-plane-wave momenta define two effective spectra:

$$\beta^u(\mathbf{q}_1; \hat{\mathbf{x}}) = \left\langle \bar{\mathcal{S}}_\ell(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{x}}) B(\mathbf{q}_1, \mathbf{q}_2; \hat{\mathbf{x}}) \right\rangle_j, \quad \beta^p(\mathbf{p}_1; \hat{\mathbf{y}}) = \left\langle \tilde{\mathcal{S}}_{\ell'}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \hat{\mathbf{y}}) B(-\mathbf{p}_1, -\mathbf{p}_2; \hat{\mathbf{y}}) \right\rangle_{j'}, \tag{45}$$

each a Legendre series in its momentum and line of sight. Then

$$\begin{aligned}
C_{\ell\ell'}^{BB,(1 | 2'3')} &= \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2} \sum_{JJ'} \\
&\times \int d^3x d^3s \omega^3(\mathbf{x}) \omega^3(\mathbf{x} + \mathbf{s}) \\
&\times \mathcal{K}_{J0}^{(i)}[\beta_J^u](\hat{\mathbf{x}}, \cdot; -\mathbf{s}) \mathcal{K}_{J'0}^{(i')}[\beta_{J'}^p](\hat{\mathbf{y}}, \cdot; -\mathbf{s}),
\end{aligned} \tag{46}$$

which uses the power-spectrum formula with weights (ω^3, ω^3) . In the box this is $\delta_{ii'}/V_i$ times the product of the two bispectra averaged with the estimator weights, the first term of eq. (32) of Sugiyama et al. (2020). It grows as B^2 , faster than any other term, which is why the Gaussian-only covariance fails at $k \gtrsim 0.1$.

10.3 PT and the super-sample overlap

Here a cross pair plus a connected quadruple. The nine cross pairs have the same coincidence table as the Gaussian pairings. For $(11')$ the pair sits at $\mathbf{y} = \mathbf{x} + \mathbf{s}$ ($R1, P(\mathbf{p}_1; \hat{\mathbf{y}})$), and the quadruple $(2, 3, 2', 3')$

at the anchor, $\omega^4(\mathbf{x})e^{i(\mathbf{q}_1-\mathbf{p}_1)\cdot\mathbf{x}}T(\mathbf{q}_2, -\mathbf{p}_2, \mathbf{p}_1 + \mathbf{p}_2; \hat{\mathbf{x}})$. The plane wave carries the beat $\mathbf{q}_1 - \mathbf{p}_1$. With A4 ($\mathbf{p}_1 \rightarrow \mathbf{q}_1$ inside T),

$$\begin{aligned}\tau(\mathbf{q}_1; \hat{\mathbf{x}}) &= \left\langle \left\langle \bar{\mathcal{S}}_\ell(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{x}}) \tilde{\mathcal{S}}_{\ell'}(\hat{\mathbf{q}}_1, \hat{\mathbf{p}}_2, \hat{\mathbf{x}}) T(\mathbf{q}_2, -\mathbf{p}_2, \mathbf{q}_1 + \mathbf{p}_2, -\mathbf{q}_1 - \mathbf{q}_2; \hat{\mathbf{x}}) \right\rangle \right\rangle_{jj'} \\ &= \sum_J \tau_J(\mathbf{q}_1) \mathcal{L}_J(\hat{\mathbf{q}}_1 \cdot \hat{\mathbf{x}}),\end{aligned}\tag{47}$$

$$\begin{aligned}C_{\ell\ell'}^{PT,(11')} &= \frac{(4\pi)^3 HH' \sqrt{NN'}}{I_3^2} \sum_{JL_1} \\ &\times \int d^3x d^3s \omega^4(\mathbf{x}) W(\mathbf{x} + \mathbf{s}) \\ &\times \mathcal{K}_{J0}^{(i)}[\tau_J](\hat{\mathbf{x}}, \cdot; -\mathbf{s}) \mathcal{K}_{0L_1}^{(i')} [P_{L_1}](\cdot, \hat{\mathbf{y}}; \mathbf{s}),\end{aligned}\tag{48}$$

i.e. weights (ω^4, W) . The box limit is the PT term of eq. (33) of Sugiyama et al. (2020). τ is a Monte Carlo over the two shells, as in `thecov.trispectrum`.

Do not double count super-sample covariance. Setting $\varepsilon = \mathbf{q}_1 - \mathbf{p}_1 = 0$ inside T (A4) drops its response terms $\propto P(\varepsilon)$. Integrated against the window, those terms **are** the super-sample covariance of the bispectrum, the analogue of T_0 's squeezed part for $P(k)$. So either use T at $\varepsilon = 0$ and add the super-sample term separately (what `thecov` does for P), or keep the full ε dependence and add nothing. Not both.

10.4 Shot noise, models, calibration

Poisson terms. For $F = w(n_g - \alpha n_r)$ the connected three-point function has:

- a one-shared-galaxy term with weight $\bar{n}w^2 \cdot \omega$, from galaxies only, because randoms do not cluster with the third field;
- a three-shared term with weight $(1 - \alpha^2)\bar{n}w^3$.

Hence $\omega^3 B \rightarrow \omega^3 B + \bar{n}w^2 \omega [P(k_a) + P(k_b) + P(k_c)] + (1 - \alpha^2)\bar{n}w^3$. The four-point function follows the same rule. The pair counts needed are those with weights built from $\omega^2, \omega^3, \omega^4, \bar{n}w^2\omega, \bar{n}w^2\omega^2, S^2$, all of which `WindowLibrary` can accumulate in one pass.

Models.

- B and T at tree level with the SCF99 kernels already in `thecov` (B in `discreteness.py` and `ssc.py`, T in `trispectrum.py`, Galileon bias), on the same damped P_{lin} as the Gaussian term, with the fingers-of-God damping on every Z_1 leg.
- β and τ need B on two-magnitude grids and T on general quadrilaterals. The kernels support this, but the current T driver integrates only parallelograms.
- P_5 (in PB) and P_6 (in BB) are corrections to corrections and are dropped.

Calibration.

- The measured multipoles can replace the model $B_{\ell''}$ in β and $\beta^{u,p}$, lightly smoothed; in noisy bins the tree model fitted to $\hat{B}$ is safer.
- T cannot be measured, but its squeezed part (position-dependent power) and its collapsed part (covariance of $\hat{P}$ across sub-volumes) can.
- One residual amplitude per term is then fitted to the mocks and checked against these data-side estimates.

11 Summary

term	window enters through	radial kernel	size on the diagonal
Gaussian class 1 (2 pairings)	$\mathcal{Q}_n(r_1, r_2)$: three-point, anchored pair sums	$\overline{jj}\mathcal{P}$, 2D	leading
Gaussian class 2 (4)	same $\mathcal{Q}_n$	$\overline{jj}\mathcal{P}^{(2)}$, 2D	$\approx \Delta k/k$
$k_3 \approx 0$ (1)	long-mode variance of $\omega^2 F$	none	on $i = j, i' = j'$; zero for uniform ω
PB	$\mathcal{Q}^{\omega^3 W}(s)$	power-spectrum kernel with $P\beta$	first order in B
BB	$\mathcal{Q}^{\omega^3 \omega^3}(s)$	same, with β^u, β^p	grows as B^2
PT	$\mathcal{Q}^{\omega^4 W}(s)$	same, with τ	plus separate SSC

12 What changed relative to `bispectrum_covariance.typ`

The physics and the results are the same. These are the differences:

1. $H_{202} = +1/\sqrt{5}$ (in general $H_{\ell 0 \ell} = (-1)^\ell / \sqrt{2\ell + 1}$). This is what makes $\mathcal{S}_{202} = \mathcal{L}_2$; the old sign gave $-\mathcal{L}_2$. [status (i)]
2. All harmonics at the five window directions are unconjugated, the invariant basis Φ_n is orthonormal for that variance, and T_n is defined as a projection. This removes the mismatch between T and E . [status (ii)]
3. K_1, K_2 run over all integers in their triangles, not same-parity sets.
4. The box value of the window invariants is stated: $\mathcal{Q}_n^{\text{box}} = V\sqrt{N/4\pi H}\delta_{\lambda_1 0}\delta_{\lambda_2 0}$. [status (iii)]
5. Class 2 is written in the variables $(\mathbf{p}_1, \mathbf{p}_3)$, one primed momentum per leg, so it shares the window functions $\mathcal{Q}_n$ with class 1. [status (iv)]
6. The rank bound reads $a \leq \lambda_w + \ell_1 + \ell'_1 + L_1$.
7. The $k_3 \approx 0$ term $(12)(1'2')(33')$ is included as a seventh Gaussian term. [status (v)]
8. Self-pair subtraction is added and the cost corrected to $\sim 10^{12}$. [status (vi)]
9. BB :
 - The displayed pairing was $(1 | 1'3')$ (coincidence $k_1 \approx k'_2$), not $(1 | 2'3')$; the derivation above uses $(1 | 2'3')$.
 - On the diagonal with $i \neq j$ only two of the four leg-leg pairings survive.
 - The same caveat applies to the second class-1 pairing.
10. The one-shared-galaxy Poisson weight of the three-point function is $\bar{n}w^2\omega$, without the factor $(1 + \alpha)$.
11. The non-Gaussian terms are reorganised: averaging the non-plane-wave momenta first turns each into the power-spectrum formula with an effective spectrum. Approximation A4 is stated explicitly.
12. New: the real-space B_{000} warm-up Equation 18, Tools 2–3 (with both signs of Tool 3 verified numerically), and the explicit class-2 suppression count ($\approx \Delta k/k$ overall).